1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Elevated circulating CK-MM, as a marker of subclinical skeletal muscle injury, reflects a systemic inflammatory state that increases vascular permeability in bone marrow microvasculature. This enhanced permeability facilitates hematogenous dissemination and extravasation of circulating prostate cancer cells, promoting bone colonization. Once in bone, disseminated cells drive osteoblastic remodeling through BMP/TGF-β signaling, leading to predominantly osteoblastic bone metastases. The effect is predicted to be dose- and time-dependent and may be mitigated by interventions that reduce systemic inflammation, stabilize endothelial junctions, or inhibit BMP signaling.

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: . (b) In vivo mouse model of prostate cancer metastasis with muscle injury to elevate CK-MM; monitor CK-MM levels, vascular permeability in bone marrow (Evans blue or similar), and metastatic burden with bioluminescence and micro-CT; histology for osteoblastic markers (osteocalcin, RUNX

Prediction 2: . (c) Interventions: anti-inflammatory (e.g., IL-6 axis inhibition), endothelial-stabilizing approaches, and BMP pathway inhibitors to test rescue of the metastatic phenotype. Data interpretation should consider CK-MM as a biomarker reflecting systemic inflammation rather than a direct causal driver.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions